

Comparative Study of Regression Algorithms for Crop Price Prediction: A Case Study of Nepal's Kalimati Market

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Abstract—Agricultural commodity prices in Nepal fluctuate sharply due to seasonal supply changes, climatic disruptions, and import dependency, making it difficult for farmers and traders to plan production and sales. This study compares six regression algorithms, Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regressor, Support Vector Regression (SVR), and a Multi-Layer Perceptron (MLP) Regressor, for predicting the daily average price of three major vegetables (Tomato, Potato, and Onion) traded at Nepal's largest wholesale market, Kalimati Tarkari Bazaar. Using ten years of historical price data (2013–2022) from a public Kaggle dataset, the study builds lag-based and seasonal features, then evaluates each model using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Linear Regression achieved the best performance ($R^2 = 0.9186$, $MAE = 3.97$ NPR/Kg), substantially outperforming a prior Prophet-based study on the same dataset ($R^2 = 0.42$). A dedicated comparative experiment further shows that feature scaling has negligible effect on linear and tree-based models but provides a modest improvement for SVR. These findings suggest that simple, interpretable regression models with well-engineered lag features can provide highly accurate, low-cost price forecasting tools for Nepali agricultural markets.

Keywords—crop price prediction, feature scaling, Kalimati market, machine learning, regression algorithms

I. INTRODUCTION

Agriculture accounts for roughly 24–25% of Nepal's Gross Domestic Product and employs more than 60% of the workforce [9]. That economic weight has not translated into price stability for the people who grow and trade the produce. At the Kalimati Fruits and Vegetables Market in Kathmandu, the country's largest wholesale hub, prices swing sharply from week to week. Seasonal production cycles, monsoon disruption, transport bottlenecks, and dependence on imported produce all feed into that volatility.

A farmer deciding when to harvest or sell is, in effect, guessing. The cost of guessing wrong shows up as produce left to spoil or sold at the bottom of a price cycle. Machine learning offers a practical way in: given enough price history, a regression model can learn the short-term structure in that history and turn it into a usable forecast.

This study addresses the following research question: Which regression algorithm most accurately predicts the market price of major vegetables (Tomato, Potato, and Onion) at Kalimati market based on historical price trends? A second question follows from it: does feature scaling meaningfully affect model performance? This is a design choice much of the agricultural price prediction literature passes over without comment.

The main contributions of this paper are: (1) a controlled, reproducible comparison of six regression algorithms spanning linear, tree-based, kernel-based, and neural families on a real Nepali agricultural dataset; (2) an engineered feature set incorporating lag-based price history and seasonal indicators, validated through correlation analysis and a feature-subset ablation; (3) three controlled experiments isolating the effects of feature scaling, scaler choice, and hyperparameter tuning; and (4) a direct comparison against prior time-series forecasting work on the same dataset, demonstrating substantial accuracy gains.

II. RELATED WORK

A. Nepal-Based Studies

R. Nepal et al. [9] analyzed seasonal price behavior in the Kalimati Tarkari dataset (2013–2023) using Facebook Prophet, logistic regression, and STL decomposition, finding strong seasonal patterns with peak prices in winter and lowest prices during monsoon. Their Prophet-based forecasting model achieved an average MAE of 32.95 and R^2 of 0.42. Malla [10] proposed a Kalimati Vegetable Price Index using a 14-model stacking ensemble with 64 engineered features including festival calendar effects, finding tree-based ensembles most robust while transformer architectures struggled with dataset noise. A separate exploratory study [11] examined long-term trends and seasonality in the same market using descriptive statistics without building predictive models.

B. International Studies

Several studies have applied Decision Tree Regression to crop price forecasting. Rachana et al. [3] reported 97.32% accuracy using crop name, month, year, rainfall, and wholesale price index as features, underscoring the value of time-based feature engineering. Comparative studies combining weather

data with regression models [5] and evaluating GRNN, SVR, Random Forest, and Gradient Boosting against ARIMA for brinjal price forecasting in India [8] consistently found machine learning approaches outperforming classical time-series models. Guo et al. [8] specifically studied Potato and Tomato price behavior using graph neural networks, characterizing Potato as exhibiting stable price behavior and Tomato as highly volatile. The exploratory analysis in Section VI reproduces that same pattern on Kalimati data.

C. Research Gap

Nepal-specific studies on the Kalimati market [9], [10] apply complex time-series or ensemble methods without a systematic, controlled comparison of standard regression algorithms, and none isolate the effect of feature scaling. International studies [1]-[8] similarly favor complex or single-crop approaches. This study addresses this gap through a controlled, multi-crop comparison of interpretable regression algorithms with an explicit feature-scaling experiment on Nepali market data.

III. DATASET DESCRIPTION

This study uses the publicly available Kalimati Tarkari Dataset [12], sourced from Kaggle, containing daily wholesale price records scraped from the Kalimati Fruits and Vegetables Market for the period June 2013 to April 2022. The raw dataset contains 280,862 records across 136 commodities with six columns: Commodity, Date, Unit, Minimum, Maximum, and Average price (NPR/Kg).

For this study, three major vegetables were selected: Tomato, Potato, and Onion, representing 15 sub-variety entries (e.g., “Tomato Big (Nepali)”, “Onion Dry (Indian)”) that were consolidated into three crop categories. After filtering, the dataset comprised 28,197 records. Table I summarizes the dataset composition.

TABLE I
DATASET SUMMARY BY CROP

Crop	Records	Mean (NPR/Kg)	Std. Dev.
Onion	7,280	64.07	41.63
Potato	9,965	38.16	16.71
Tomato	10,931	46.73	22.18

Table II describes each field in the raw dataset prior to feature engineering, including data type and role in the modeling pipeline.

TABLE II
RAW DATASET FIELD DESCRIPTION

Field	Type	Role
Commodity	String	Crop/sub-variety identifier
Date	Date	Price record date
Unit	String	Measurement unit (Kg)
Minimum	Float	Daily minimum price

Field	Type	Role
Maximum	Float	Daily maximum price
Average	Float	Target variable

No missing values or duplicate records remained after preprocessing (Section IV-A). The target variable for all models is the Average daily price in NPR per kilogram.

IV. METHODOLOGY

A. Data Preprocessing

The raw file needed work before any of it was usable. Dates arrived in two formats, M/D/YYYY in the older records and YYYY-MM-DD in the newer ones, so a parser that tries both was used to standardize them. The three price columns were worse: most values were plain numbers, but some carried currency text such as “Rs 250.00”, which forced the entire column to be read as text. Stripping the currency prefix and casting to float resolved this. Rows whose date or price could not be recovered were dropped, though in practice none survived to that stage; the cleaned dataset contained no missing values and no duplicates.

B. Feature Engineering

Calendar-based features (Year, Month) and a four-class Season indicator (Winter, Spring, Monsoon, Autumn, mapped to Nepal's agricultural calendar) were derived from the Date field. Critically, lag features representing the previous day's price (Price_Lag1) and the price seven days prior (Price_Lag7) were constructed to capture short-term price persistence.

The first attempt at these lag features gave a result that made no sense: the correlation between Onion prices and their own previous-day value came out at $r = 0.058$, effectively zero. The cause was structural. Several sub-varieties share a single date in the raw data, so shifting by one row did not step back one day; it stepped sideways into a different variety, comparing Onion Green against Onion Dry. Aggregating to one mean price per (Crop, Date) pair before computing the lags fixed it. Per-crop correlations then rose to 0.965 for Onion, 0.976 for Potato, and 0.958 for Tomato, which is the behaviour expected of a price series. The final dataset held 10,814 records.

C. Regression Models

Six regression algorithms spanning four distinct algorithm families were implemented using scikit-learn, which allows the comparison to test whether any particular family suits this problem rather than any single model. The linear family is represented by Linear Regression, Ridge Regression ($\alpha = 1.0$), and Lasso Regression ($\alpha = 0.1$); the tree-based family by a Decision Tree Regressor (max depth = 8); the kernel-based family by Support Vector Regression (RBF kernel, $C = 10$, $\epsilon = 0.5$); and the neural family by a Multi-Layer Perceptron Regressor with two hidden layers of 64 and 32 units, trained for up to 500 iterations. Every model that draws on randomness was

given `random_state = 42`. Categorical features (Crop, Season) were one-hot encoded, yielding a nine-dimensional feature matrix: Month, Year, Price_Lag1, Price_Lag7, and one-hot indicators for Crop and Season.

D. Comparative Experiment: Feature Scaling

To evaluate the effect of feature scaling, each model was trained twice on identical train/test splits: once on raw (unscaled) features, and once on features standardized via z-score scaling (StandardScaler, fit on training data only). This isolates the sensitivity of each algorithm family (linear, tree-based, kernel-based) to input feature magnitude.

V. EXPERIMENTAL SETUP

Because the dataset is a time series, a chronological (non-random) 80/20 train-test split was used: the earliest 80% of records (8,651 samples) formed the training set, and the most recent 20% (2,163 samples) formed the test set. This design avoids information leakage that would occur with random shuffling, since training on future prices to predict past prices would be unrealistic in a deployed forecasting scenario.

Model performance was evaluated using four standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). All experiments were implemented in Python 3.11 using pandas, NumPy, and scikit-learn. Random seeds were fixed throughout, and the chronological split makes the partition itself deterministic, so every result reported here reproduces exactly on re-execution.

VI. RESULTS

A. Exploratory Data Analysis

Fig. 1 shows the monthly average price trend for all three crops from 2013 to 2022. Onion exhibits the highest volatility, with recurring price spikes exceeding NPR 140/Kg, attributable to its dependence on cross-border imports. Potato shows the most stable trend, consistent with prior findings on Potato price stability [8].

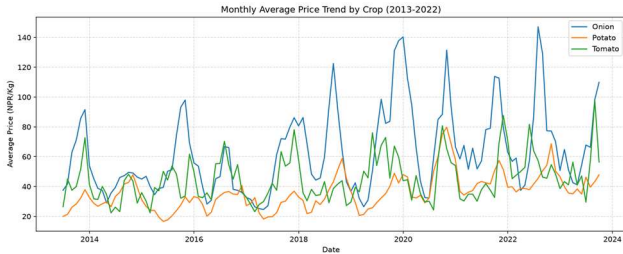


Fig. 1. Monthly average price trend by crop (2013–2022).

Fig. 2 presents seasonal price distributions. All three crops show elevated median prices and wider variance during the Monsoon season, consistent with supply disruptions from flooding and landslides, and matching the seasonal pattern independently reported by Nepal et al. [9] on the same market.

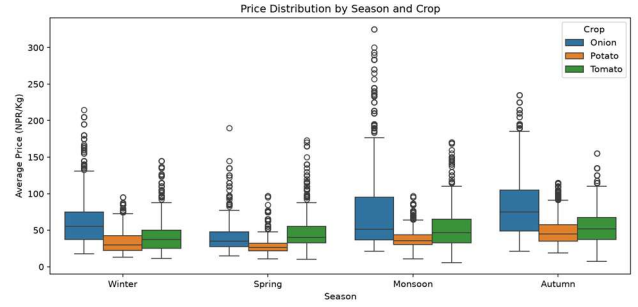


Fig. 2. Price distribution by season and crop.

Fig. 3 shows the overall price distribution for each crop. Onion displays a longer right tail extending beyond NPR 250/Kg, confirming its higher susceptibility to extreme price spikes, while Potato's distribution is tightly concentrated below NPR 100/Kg, reflecting its comparatively stable supply chain.

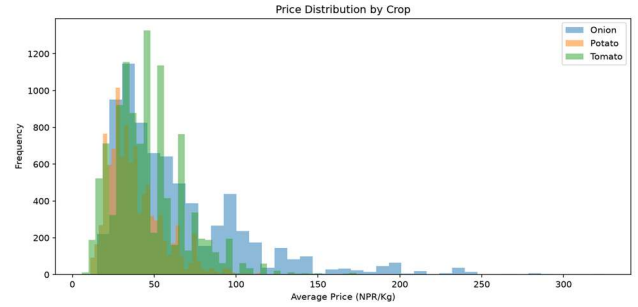


Fig. 3. Price distribution by crop (histogram).

Fig. 4 shows the correlation matrix of numeric features after the aggregation fix described in Section IV-B. Price_Lag1 and Price_Lag7 correlate strongly with each other ($r = 0.83$), reflecting short-term price persistence, and both show moderate positive correlation with the target Average price when computed globally across crops; per-crop correlations (Section IV-B) are substantially higher.

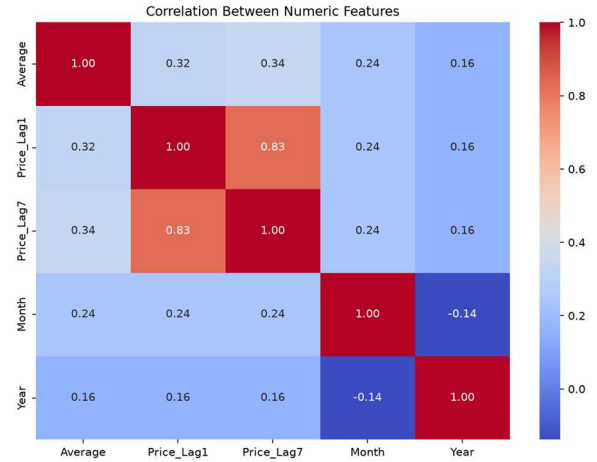


Fig. 4. Correlation between numeric features.

B. Model Comparison

Table III summarizes the performance of all six regression models on the held-out test set.

TABLE III
MODEL PERFORMANCE COMPARISON

Model	MAE	MSE	RMSE	R ²
Linear Regression	3.965	51.547	7.180	0.9186
Ridge Regression	3.966	51.543	7.179	0.9186
Lasso Regression	3.934	51.679	7.189	0.9184
Decision Tree Regressor	4.496	65.280	8.080	0.8969
SVR	4.875	83.179	9.120	0.8686
MLP Regressor	4.622	58.212	7.630	0.9081

All three linear models cluster at the top. Linear and Ridge Regression finish level at $R^2 = 0.9186$, and Lasso sits a fraction below at 0.9184 while posting the lowest MAE of the six (3.934). That split between metrics is informative: MAE weights every error equally, whereas MSE and R^2 punish large errors disproportionately, so Lasso trims typical errors slightly at the cost of doing marginally worse on the rare price spikes. Ridge adds a regularization penalty to guard against overfitting, and the fact that it changes almost nothing indicates there was little to guard against.

The more interesting result sits below them. The MLP Regressor, by far the most flexible model in the set, reaches $R^2 = 0.9081$, ahead of both the Decision Tree and SVR but still short of a plain least-squares fit. A network with two hidden layers can represent any smooth function given enough data, so its failure to beat Linear Regression is not a limitation of the model but a statement about the problem: there is little non-linear structure left to find once the lag features are in place. The same reading explains the Decision Tree and SVR results. Added capacity did not help on this task.

C. Effect of Feature Scaling

Fig. 5 and Table IV compare each model's R^2 score with and without feature standardization.



Fig. 5. Effect of feature scaling on model performance (R^2).

TABLE IV
EFFECT OF FEATURE SCALING (R^2)

Model	No Scaling	Scaled	Δ
Linear Regression	0.9186	0.9186	0.0000
Ridge Regression	0.9186	0.9186	0.0000
Lasso Regression	0.9180	0.9184	+0.0003
Decision Tree Regressor	0.8969	0.8969	0.0000
SVR	0.8658	0.8686	+0.0028
MLP Regressor	0.9120	0.9081	-0.0039

The linear and tree-based models are untouched by scaling, exactly as theory predicts. Linear, Ridge, and Lasso Regression absorb the change into their coefficients, and a decision tree splits on thresholds, so rescaling an axis moves the split point without moving the boundary. SVR is the model that should care, since it works from distances between points, and it does improve, though only by 0.0028 under standardization. The MLP moved the other way by 0.0039, which is small enough to fall within the run-to-run variation of random weight initialization and should not be read as a real effect. Section VI-E returns to SVR with a wider set of scalers and finds the choice matters more than this table suggests.

D. Prediction Accuracy Analysis

Fig. 6 plots predicted against actual prices for the best-performing model (Linear Regression). Points cluster tightly around the ideal prediction line at lower price ranges (25–100 NPR/Kg), with modestly increased scatter at higher, rarer price points (150–200 NPR/Kg), corresponding to extreme Onion price spikes.

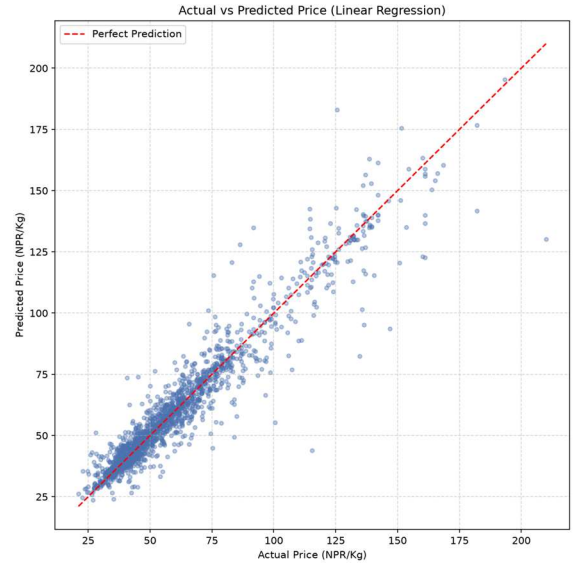


Fig. 6. Actual vs. predicted price (Linear Regression).

Fig. 7 shows the residual plot. Residuals are randomly distributed around zero with no strong systematic pattern, indicating the absence of major model bias. A mild increase in residual spread at higher predicted prices (heteroscedasticity) suggests marginally reduced precision for extreme, low-frequency price events.

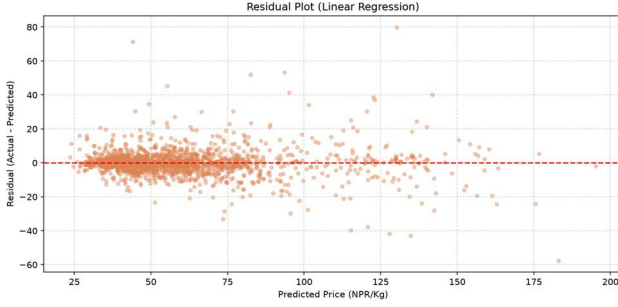


Fig. 7. Residual plot (Linear Regression).

E. Effect of Scaler Choice

Table V extends the scaling experiment by testing four preprocessing options on SVR, the model most sensitive to feature scale.

TABLE V
EFFECT OF SCALER CHOICE ON SVR

Scaler	MAE	R ²
None (raw features)	5.595	0.8658
StandardScaler	4.875	0.8686
MinMaxScaler	4.638	0.8908
RobustScaler	4.686	0.8786

This result qualifies the conclusion of Section VI-C. Standardization lifts SVR only slightly, but MinMaxScaler lifts it from $R^2 = 0.8658$ to 0.8908 , a gain roughly eight times larger. The difference lies in what each transform does with the one-hot indicator columns: MinMaxScaler maps every feature onto $[0, 1]$, leaving the binary indicators unchanged and placing the lag prices on the same footing, whereas StandardScaler rescales the sparse binary columns by their own small standard deviations and inflates them. The practical lesson is that reporting “with and without scaling” understates the question; which scaler is used matters in its own right.

F. Feature Subset Ablation

Table VI reports Linear Regression trained on three feature subsets, isolating how much each group contributes.

TABLE VI
EFFECT OF FEATURE SUBSET ON LINEAR REGRESSION

Feature Subset	MAE	R ²
Calendar only (Month, Year, Crop, Season)	14.490	0.3358
Lag only (Price_Lag1, Price_Lag7)	3.938	0.9184
All features	3.965	0.9186

The ablation settles a question the correlation analysis could only suggest. Calendar and crop identity alone explain barely a third of the variance ($R^2 = 0.3358$), while the two lag features on their own reach 0.9184 , within 0.0002 of the full model.

Almost all predictive power comes from recent price history; the calendar features contribute a negligible amount on top. This is consistent with the nature of the task, since a one-day forecast depends far more on yesterday’s market than on the month of the year.

G. Hyperparameter Tuning

Table VII reports grid search over the main hyperparameters of four models. Because the data is a time series, TimeSeriesSplit with three folds was used rather than standard k-fold, so that each validation fold always falls after its training fold.

TABLE VII
HYPERPARAMETER TUNING RESULTS

Model	Best Parameters	R ² Def.	R ² Tuned
Ridge Regression	$\alpha = 0.01$	0.9186	0.9186
Lasso Regression	$\alpha = 0.01$	0.9184	0.9186
Decision Tree	depth 6, leaf 10	0.8969	0.9133
SVR	$C = 100, \epsilon = 0.1$	0.8686	0.8708

Tuning changed almost nothing for the linear models, which is itself a finding: their default settings were already near optimal, and the search selected the weakest regularization on offer ($\alpha = 0.01$) in both cases, confirming that the problem does not call for shrinkage. The Decision Tree is the one model that gained materially, climbing from 0.8969 to 0.9133 . The search chose a shallower tree (depth 6 rather than 8) with a minimum of ten samples per leaf, indicating that the default configuration had been overfitting. Even tuned, however, it remains below every linear model.

VII. DISCUSSION

That Linear and Ridge Regression reach $R^2 = 0.9186$ says something specific: nearly all of the predictable movement in next-day Kalimati prices is already contained in the recent price history. The comparison with Nepal et al. [9] sharpens the point. Working from the same underlying dataset with Prophet, they report $R^2 = 0.42$. The gap is not evidence that Prophet is a weaker tool in general; it reflects a difference in target. Prophet decomposes a series into trend and seasonal components, which suits medium-term questions. For a one-day horizon, a plain regression fed the right lag features does more with less.

The three controlled experiments point the same way. Scaling is irrelevant for linear and tree-based models, but for SVR the choice of scaler is worth more than the decision to scale at all: MinMaxScaler adds 0.025 to R^2 where StandardScaler adds 0.003 . Hyperparameter tuning helped only the Decision Tree, and only by correcting overfitting rather than by finding hidden capability. The feature ablation is the most direct of the three: the lag features alone reach $R^2 = 0.9184$ against 0.9186 for the full set, so the calendar features are close to redundant for a one-day horizon. Together these results argue

that effort spent on feature construction returns far more here than effort spent on model selection or tuning.

A. Limitations

This study is limited to three vegetables and does not incorporate external variables such as rainfall, transportation cost, or festival calendars, which prior work [10] has shown to be informative. The observed heteroscedasticity indicates reduced reliability for extreme price events, which are precisely the events of greatest concern to farmers. Additionally, the dataset extends only to April 2022 and does not reflect more recent market conditions.

B. Practical Implications

The accuracy achieved by a simple, interpretable Linear Regression model ($MAE \approx 4$ NPR/Kg) has direct practical value: unlike opaque deep learning approaches, its coefficients can be readily inspected to understand which factors drive price changes, making it more suitable for deployment in low-resource settings such as cooperative extension offices or mobile advisory services aimed at smallholder farmers in Nepal. The low computational cost of training also means the model can be retrained frequently as new daily price data becomes available, keeping forecasts current without requiring specialized hardware.

C. Future Work

Future extensions of this work should incorporate exogenous variables such as rainfall records, temperature, and festival-calendar indicators, which prior studies [5], [10] have shown to improve forecast accuracy. Extending the crop set beyond Tomato, Potato, and Onion, and updating the dataset beyond 2022, would improve the generalizability of the findings. Finally, applying quantile regression or prediction intervals could directly address the heteroscedasticity observed in Section VI-D, providing farmers with a measure of forecast uncertainty rather than a single point estimate.

VIII. CONCLUSION

Six regression algorithms were compared on ten years of Kalimati price data for Tomato, Potato, and Onion. Linear and Ridge Regression came out ahead at $R^2 = 0.9186$, with a mean absolute error near 4 NPR per kilogram, and neither a kernel method, a decision tree, nor a two-layer neural network could better them. The same figure sits well above the $R^2 = 0.42$ reported for a Prophet-based approach on this dataset, though the two target different forecast horizons. Three further experiments sharpened the picture: scaling is irrelevant to linear and tree-based models but the scaler chosen changes SVR appreciably; tuning corrected overfitting in the Decision Tree without lifting it past the linear models; and a feature ablation showed the lag features alone carry essentially all of the accuracy. Taken together, the case here is for the simpler tool. A model this small and this interpretable can be retrained daily on a modest machine, which matters more in Nepali market

advisory settings than a marginal accuracy gain from a heavier architecture.

ACKNOWLEDGMENT

The author thanks the course instructor for guidance on research methodology and the developers of the open-source Kalimati Tarkari Dataset for making Nepali agricultural market data publicly available for research purposes.

AI USAGE STATEMENT

An AI assistant (Claude, Anthropic) was used throughout this project. Its role covered debugging the Python preprocessing pipeline, including diagnosis of the date-parsing inconsistency and the sub-variety aggregation flaw described in Section IV-B; drafting and refining the language of this manuscript from the author's results and notes; and guidance on IEEE formatting. The choice of problem and dataset, all preprocessing and modelling decisions, execution of every experiment, and the interpretation of results are the author's own. All figures and tables were generated by the author's code.

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